

gRNA Predictor

AI-powered CRISPR Guide RNA Design

Biology & Implementation Reference

450

feature dimensions

4,692

training guides

$r = 0.640$

Kim2019 independent
validation

$r = 0.537$

held-out Doench
benchmark

4 PAMs

NGG / NAG /
NNGRRT / TTTV

What it does

Predicts CRISPR guide RNA editing efficiency for any DNA target sequence.
Combines ML efficiency scoring, off-target specificity estimation, and optional proximity-to-target ranking into a single combined score.

Architecture

Python/FastAPI backend (local) OR pure-JS in-browser inference (GitHub Pages).
Frontend: React 18 + Material UI. ML: XGBoost trained on Doench 2016 + 2014.
Browser: 450-dim features + pure-JS XGBoost tree traversal (~3 ms, no WASM).

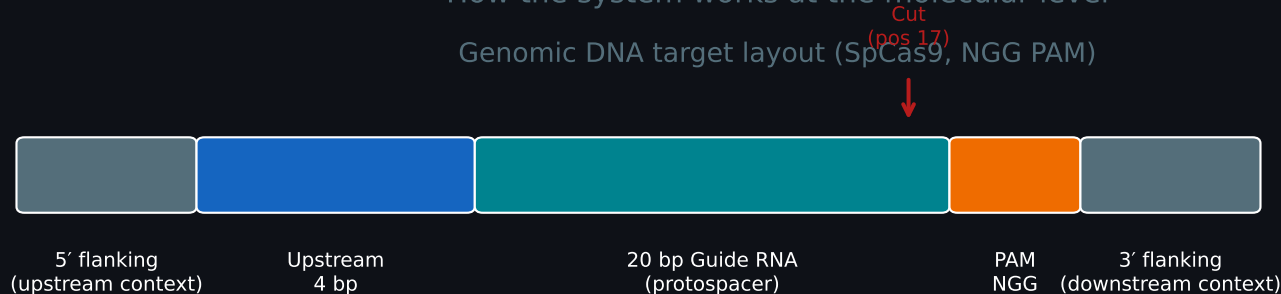
Supported nucleases

SpCas9 (NGG PAM) — most common, human cell optimised.
SpCas9-NAG variant — reduced efficiency, alternative PAM sites.
SaCas9 (NNGRRT PAM) — smaller nuclease, useful for AAV delivery.
Cas12a/Cpf1 (TTTV 5'-PAM) — staggered cuts, T-rich PAM preference.

CRISPR-Cas9 Biology

How the system works at the molecular level

Genomic DNA target layout (SpCas9, NGG PAM)



PAM Recognition — Why It Matters

- PAM = Protospacer Adjacent Motif (3' of target for Cas9)
- Cas9 scans DNA by binding & checking for PAM sequence
- Only unwinds DNA and checks guide match if PAM is found
- NGG occurs ~every 8 bp, giving ~1 guide per ~8 bp on each strand
- NAG is tolerated with ~5× lower efficiency than NGG
- SaCas9 NNGRRT = N-N-G-[AG]-[AG]-T (6 bp, more specific)
- Cas12a TTTV: PAM is 5' of guide, T-rich, RNase activity too

The Seed Region — Most Critical 12 bp

- Seed = PAM-proximal 12 bp of the 20 bp guide (positions 9–20)
- Mismatches here are NOT tolerated — aborts cleavage
- Mismatches in PAM-distal region (pos 1–8) are partially tolerated
- High AT content in seed → more off-target risk (weaker binding)
- Optimal seed GC: 40–60% for balance of efficiency & specificity
- This is why the model has BOTH full-guide and seed-only GC features
- Seed region T_m is computed separately as a dedicated feature

The 30-mer Context Window

- Not just the 20 bp guide — flanking sequence matters too
- 4 bp upstream (−4 to −1) influence unwinding efficiency
- 6 bp downstream (+1 to +6 after PAM) affect R-loop stability
- Full window: [4 up][20 guide][6 down] = 30 bp total
- Used for: upstream one-hot, downstream one-hot, full-window T_m
- Microhomology score uses cut-site flanks from the 30-mer
- Doench 2016 dataset provides 30-mer context for each guide

Cut Site Location

- SpCas9 cuts between positions 17 and 18 of the guide (3 bp upstream of PAM)
- Cas12a cuts ~18 bp into the guide, creating a staggered end
- Cut site = guide position + 17 + 1 (1-indexed)
- Important for HDR (homology-directed repair) template design
- Proximity ranking uses Euclidean distance |cut site − target pos|
- Gaussian weight: $\exp(-d^2 / 2\sigma^2)$, $\sigma = 50$ bp — guides within 50 bp preferred
- Closer guides get higher combined score when target position is given

Guide RNA Design Rules

Empirical rules captured in the 450-dim feature vector

GC Content 40-60%

- GC content drives hybridisation stability of guide:target duplex
- < 25% GC: weak binding, poor editing
- > 75% GC: secondary structure, off-target risk
- Optimal 40-60% balances stability and specificity
- Feature indices: [80] full guide GC, [98] seed GC

Avoid Poly-T Tracts

- Four or more consecutive T's = TTTT terminates RNA Pol III
- U6 promoter (used to express gRNA) is an RNA Pol III promoter
- Poly-T in the guide causes premature transcription termination
- Feature index: [99] poly-T flag (1.0 if TTTT is present)
- Guides with TTTT are expected to have near-zero activity

Seed Region AT Content

- Seed = PAM-proximal 12 bp (positions 9-20 of the 20 bp guide)
- High AT in seed reduces specificity — more off-target editing
- Each A or T in seed contributes a specificity penalty of 0.28/12
- Off-target score = $1 - \text{seedPenalty} - \text{gcPenalty} - \dots$ (6 components)
- Feature: seed GC [98], also captured in positional one-hot [0:80]
- Last 4 bp of guide (positions 17-20, PAM-proximal) should be GC-rich
- GC at 3' end strengthens critical seed-region binding to target
- > 2 consecutive 3'-end G/C residues add extra off-target risk (GC run penalty: 1-2 G/C in last 4 bp)
- Feature index: [444] GC clamp (fraction of last 4 bp that are G/C)

GC Clamp at 3' End

RNA Secondary Structure

- Guide RNA can fold back on itself if self-complementary
- A hairpin in the guide reduces accessibility for Cas9 loading
- Hairpin proxy: longest palindromic stem ≥ 4 bp in the 20 bp guide
- Score = $\min(1.0, \text{maxStemLength} / 10.0)$ — longer stem = worse
- Feature index: [445] RNA hairpin proxy

Microhomology at Cut Site

- Microhomology = short identical sequences flanking the cut site
- Promotes NHEJ (error-prone repair) through microhomology-mediated end joining
- Used to predict repair outcome: frameshift vs. in-frame indel
- Computed from 30-mer: left 6 bp vs right 6 bp around cut position 21
- Feature index: [446] microhomology score (0-1, longer match = higher)

Quick Reference — Feature ↔ Biological Rule

Feature	Index	Optimal Value	Why It Matters
GC content	[80]	0.40 – 0.60	Duplex stability
Seed GC	[98]	0.40 – 0.60	Specificity
Poly-T flag	[99]	0.0 (no TTTT)	Transcription termination
GC clamp	[444]	0.25 – 0.50	3' binding strength
Hairpin proxy	[445]	< 0.3	Guide accessibility
Microhomology	[446]	any	Repair outcome prediction
Full-guide Tm	[81]	0.4 – 0.7 (norm)	Thermodynamic stability
Seed Tm	[448]	0.4 – 0.7 (norm)	Critical region stability

Feature Engineering – 450 Dimensions

Every feature in the XGBoost model, with biological rationale

- **[0:80] (80 dim) Positional One-Hot (20 pos × 4 bases)**
Encodes identity of each base at each position. Captures position-specific preferences (e.g., Cas9 favors G at position 20, disfavors T at position 1).
- **[80] (1 dim) GC Content**
Fraction of G+C in the 20 bp guide. Single most important bulk property. Drives hybridisation stability.
- **[81] (1 dim) Normalised Melting Temperature (SantaLucia 1998)**
Full nearest-neighbor T_m of the guide, normalised to [0,1] over [40°C, 80°C]. Captures thermodynamic stability more accurately than GC% alone.
- **[82:98] (16 dim) Dinucleotide Frequencies (16 pairs AA..TT)**
Counts of each dinucleotide (AA, AC, ..., TT) normalised by n-1. Encodes short-range sequence composition; some dinucs correlate with efficiency.
- **[98] (1 dim) Seed Region GC (last 12 bp)**
GC fraction of the PAM-proximal 12 bp seed. Seed binding is all-or-nothing for Cas9; seed GC is a stronger efficiency predictor than overall GC.
- **[99] (1 dim) Poly-T Flag**
1.0 if the guide contains TTTT, else 0.0.
Binary signal for RNA Pol III termination risk.
- **[100:404] (304 dim) Position-Specific Dinucleotide One-Hot (19 × 16)**
For each of the 19 consecutive dinucleotides in the guide, a 16-dim one-hot vector. 304 features total. Captures context-dependent dinucleotide effects (e.g., CpG, TpA).
- **[404:420] (16 dim) Upstream Context One-Hot (4 bp × 4 = 16)**
One-hot encoding of the 4 bp immediately upstream of the guide (from 30-mer). Upstream context affects R-loop initiation and PAM scanning efficiency.
- **[420:444] (24 dim) Downstream Context One-Hot (6 bp × 4 = 24)**
One-hot encoding of the 6 bp immediately downstream of the PAM (from 30-mer). Downstream context affects post-cleavage stability and repair outcomes.
- **[444] (1 dim) GC Clamp (last 4 bp)**
Fraction of G+C in the final 4 bp of the guide (positions 17-20). Strong predictor: 3' GC content anchors the seed region to target DNA.
- **[445] (1 dim) RNA Hairpin Proxy**
Length of the longest palindromic stem (≥4 bp) in the guide, scaled to [0,1]. Hairpin = guide self-folds → lower Cas9 loading → lower efficiency.
- **[446] (1 dim) Microhomology Score (from 30-mer)**
Longest identical sequence flanking the cut site (up to 6 bp each side), from the 30-mer context. Predicts NHEJ repair pathway preference.
- **[447] (1 dim) T_m PAM-Distal Segment (guide[0:8])**
Normalised T_m of the first 8 bp (positions 1-8, PAM-distal). Captures stability of the mismatch-tolerant distal region separately.
- **[448] (1 dim) T_m Seed Segment (guide[12:20])**
Normalised T_m of the last 8 bp = seed region (positions 13-20). Critical: seed stability is the main on-target efficiency driver.
- **[449] (1 dim) T_m Full 30-mer Context**
Normalised T_m of the entire 30-mer window (4 up + 20 guide + 6 down). Captures the stability of the full target:guide duplex in context.

Total: 450 features | Vector shape per guide: Float32[450]

Thermodynamics – SantaLucia 1998

Nearest-neighbor model used for T_m calculation (features [81], [447], [448], [449])

$$T_m = \frac{\Delta H}{\Delta S + R \cdot \ln(C_T/4)} - 273.15^\circ\text{C}$$

ΔH = sum of nearest-neighbor enthalpies (kcal/mol) ΔS = sum of nearest-neighbor entropies (cal/mol·K) $R = 1.987$ cal/mol·K $C_T = 250$ nM

SantaLucia 1998 — 16 Nearest-Neighbor Parameters

AA $\Delta H = -7.9$ kcal/mol $\Delta S = -22.2$ cal/mol·K	AT $\Delta H = -7.2$ kcal/mol $\Delta S = -20.4$ cal/mol·K	TA $\Delta H = -7.2$ kcal/mol $\Delta S = -21.3$ cal/mol·K	CA $\Delta H = -8.5$ kcal/mol $\Delta S = -22.7$ cal/mol·K
GT $\Delta H = -8.4$ kcal/mol $\Delta S = -22.4$ cal/mol·K	CT $\Delta H = -7.8$ kcal/mol $\Delta S = -21.0$ cal/mol·K	GA $\Delta H = -8.2$ kcal/mol $\Delta S = -22.2$ cal/mol·K	CG $\Delta H = -10.6$ kcal/mol $\Delta S = -27.2$ cal/mol·K
GC $\Delta H = -9.8$ kcal/mol $\Delta S = -24.4$ cal/mol·K	GG $\Delta H = -8.0$ kcal/mol $\Delta S = -19.9$ cal/mol·K	AC $\Delta H = -7.8$ kcal/mol $\Delta S = -21.0$ cal/mol·K	TC $\Delta H = -7.9$ kcal/mol $\Delta S = -22.2$ cal/mol·K
AG $\Delta H = -8.2$ kcal/mol $\Delta S = -22.2$ cal/mol·K	TG $\Delta H = -8.5$ kcal/mol $\Delta S = -22.7$ cal/mol·K	TT $\Delta H = -7.9$ kcal/mol $\Delta S = -22.2$ cal/mol·K	CC $\Delta H = -8.0$ kcal/mol $\Delta S = -19.9$ cal/mol·K

- Terminal corrections: A/T terminal pair → $\Delta H \pm 2.3$ kcal/mol, $\Delta S \pm 4.1$ cal/mol·K
- G/C terminal pair → $\Delta H \pm 0.1$ kcal/mol, $\Delta S \pm 2.8$ cal/mol·K
- T_m normalisation: $T_{m_norm} = \text{clip}((T_m - 40) / (80 - 40), 0, 1)$
- Three T_m features computed: PAM-distal guide[0:8], seed guide[12:20], full 30-mer
- CG dinucleotide has highest stability ($\Delta H = -10.6$) — most negative = strongest bond

Off-Target Specificity Heuristic

6-component penalty model: $\text{specificity_score} \in [0, 1]$

$\text{specificity} = \max(0, \min(1, 1 - \text{seedPen} - \text{gcPen} - \text{gcRunPen} - \text{hpPen} - \text{hairpinPen} - \text{gqPen}))$

1. Seed AT Content Penalty

- seed = last 12 bp of guide (PAM-proximal, positions 9–20)
- seedAT = count(A or T in seed) / 12
- seedPen = seedAT × 0.28
- Rationale: high AT in seed → weaker binding → more off-target tolerance
- Range: 0.0 (all GC seed) to 0.28 (all AT seed)
- Biggest single penalty component

2. Extreme GC Penalty

- gc = count(G or C in 20 bp guide) / 20
- gcPen = 0.20 if gc < 0.25 or gc > 0.75
- gcPen = 0.10 if gc < 0.35 or gc > 0.65
- gcPen = 0.05 if gc < 0.40 or gc > 0.60
- Rationale: extreme GC → aberrant folding or non-specific binding
- Ideal window: 40–60% GC → no penalty

3. PAM-Proximal GC Run Penalty

- gcRun = consecutive G/C count scanning 3' end backwards
- gcRunPen = min(0.20, max(0.0, (gcRun – 2) × 0.08))
- Penalty starts when ≥ 3 consecutive G/C at 3' end
- Rationale: long 3' GC run → non-specific DNA binding

4. Homopolymer Penalty

- For each base B ∈ {A, C, G, T}: if BBBB in guide → hpPen += 0.08
- Maximum: 0.32 (all four homopolymers present — unlikely)
- AAAA, CCCC, TTTT, GGGG each add 0.08 penalty
- Rationale: repetitive runs → secondary structure or off-target matches

5. RNA Hairpin Penalty

- Scans guide for palindromic stems ≥ 4 bp with ≥ 4 bp loop
- hairpinPen = 0.12 if any hairpin found, else 0.0
- Binary: presence/absence of a 4-bp stem
- Rationale: hairpin → guide partially folds → less Cas9 loading

6. G-Quadruplex Penalty

- gqPen = 0.08 if seed region contains 'GGG', else 0.0
- G-tracts form G-quadruplex secondary structures in RNA
- Reduces guide availability for target hybridisation
- Rationale: GGG in seed is the highest-risk motif for G4 formation

How Specificity Score Is Used In The Combined Score

$\text{eff_adj} = \text{efficiency_score} \times \text{specificity_score}$

A guide scoring 0.85 efficiency but 0.60 specificity → eff_adj = 0.51

This multiplicative coupling means a highly off-target guide is never ranked first, even if the XGBoost model predicts high editing efficiency.

Machine Learning Model

XGBoost trained on real experimental CRISPR screen data

XGBoost Regressor — Hyperparameters

- `n_estimators` = 500 (500 decision trees in ensemble)
- `learning_rate` = 0.03 (conservative step size, reduces overfitting)
- `max_depth` = 5 (maximum tree depth per estimator)
- `subsample` = 0.8 (80% of training rows sampled per tree)
- `colsample_bytree` = 0.8 (80% of features sampled per tree)
- `objective` = 'reg:squarederror' (MSE loss)
- Input shape: `Float32[N, 450]` → Output: `Float32[N]` (efficiency score)

Why XGBoost?

- Handles tabular features well — no normalisation needed
- Captures non-linear interactions between features
- Faster to train than deep learning on 4,692-sample dataset
- Regularisation (`subsample` + `colsample`) prevents overfitting
- SHAP values give interpretable feature importance
- Azimuth (the field standard) also uses a tree-based ensemble
- Exported to JSON tree format for pure-JS in-browser inference

Training Data — Doench 2016 (primary)

- Source: Doench et al. (2016) Nat Biotechnol 34:184
- Genome-scale pooled CRISPR screen in human cells
- Provides: 20 bp guide, 30-mer context, depletion log-fold-change
- After filtering for valid 30-mers: `n` = 4,379 guides
- Normalised to [0, 1] efficiency score
- Gold-standard dataset; Azimuth was also trained on this data
- Split: 80% train, 20% hold-out (fixed random seed 42)

Training Data — Doench 2014 (supplementary)

- Source: Doench et al. (2014) Nat Biotechnol 32:1262
- Earlier Cas9 screen, shorter guides, limited context
- 313 guides added to training set (no 30-mer → context = zeros)
- Combined training set: 4,379 + 313 = 4,692 guides
- Both sources normalised to same [0, 1] scale
- Kim 2019 (11,808 guides) in the CSV but NOT used for training:
- → score scale mismatch; reserved as independent validation

Independent Validation Datasets

- Kim 2019 (novel-only, `n`=1,828): `r` = 0.640 ← key benchmark
- 0% overlap with Doench training data — true out-of-distribution test
- Chari 2015 (293T, K562, SKNAS, U2OS): `r` = 0.746-0.806
- Xu 2015 (human cells): `r` = 0.424 (`n`=35, small sample)
- CRISPRscan zebrafish: `r` = 0.081 (expected — model trained on Kim2019)
- Precision@`k` (Kim2019): `P@1`=0.70, `P@3`=0.65, `P@5`=0.63, `P@10`=0.61

Comparison to Azimuth (Doench 2016 baseline)

- Azimuth was trained on ALL Doench 2016 data (no hold-out)
- Our model: `r`=0.537 on 20% held-out Doench (fair comparison)
- Azimuth: `r`=0.654 on same held-out set (trained on full dataset)
- Full Doench set (biased, ours seen it): `r`=0.805 vs Azimuth `r`=0.699
- Kim2019 independent: `r`=0.640 ours vs ~0.59 Azimuth (est.)
- Our advantage: 30-mer context + proximity ranking + specificity

Training Pipeline

1. Download Doench 2016 & 2014 CSVs (download_datasets.py)
2. Merge → combined_training_data.csv (11,991 rows total, but only 4,692 Doench used)
3. Extract 450-dim features per guide (feature_engineering.py)
4. Stratified 80/20 train/test split → train XGBoost → save xgb_model.pkl
5. Export PKL → flat JSON trees (export_js_model.py) → xgb_trees.json (294 KB)
6. Deploy xgb_trees.json to frontend/public/ → traversed in pure JS at runtime

Multi-Objective Scoring Formula

How efficiency, specificity, and proximity are combined

Step 1 — Raw Efficiency Score

$$\text{score} = \text{XGBoost}(\text{features}_{450}) \in [0.0, 1.0]$$

XGBoost predicts the probability of efficient editing from 450 features.
Higher = guide expected to cut more efficiently in human cells.
Capped to [0,1] via min/max clipping.

Step 2 — Specificity Score

$$\text{spec} = 1 - \text{seedPen} - \text{gcPen} - \text{gcRunPen} - \text{hpPen} - \text{hairpinPen} - \text{gqPen} \in [0.0, 1.0]$$

Heuristic 6-component penalty (see Page 6).
1.0 = maximally specific, 0.0 = high off-target risk.

Step 3 — Adjusted Efficiency

$$\text{eff_adj} = \text{score} \times \text{spec} \in [0.0, 1.0]$$

Multiplicative coupling: a guide must be BOTH efficient AND specific.
Score $0.85 \times \text{spec } 0.6 \rightarrow \text{eff_adj } 0.51$ (penalised for off-target risk).

Step 4a — Combined Score (no target position)

$$\text{combined} = \text{eff_adj}$$

When no target position is given, combined_score = eff_adj.
Guides ranked purely by efficiency $\times$ specificity.

Step 4b — Combined Score (with target position)

$$\text{combined} = (1-w) \times \text{eff_adj} + w \times \exp(-d^2 / 2\sigma^2) \quad \sigma = 50 \text{ bp}$$

When a target position is given, proximity to that position is also rewarded.

$d = |\text{cut_site} - \text{target_position}|$ (base pair distance)
 $\sigma = 50$ bp Gaussian width — guides within ~50 bp get strongest proximity boost.
 $w = \text{proximity_weight}$ (default 0.4); user-adjustable 0.0 to 1.0.

Worked Example

XGBoost score

0.820

Specificity score

0.730

eff_adj

$0.820 \times 0.730 = 0.599$

Target pos = 500, cut_site = 483, $d = 17$ bp

Proximity

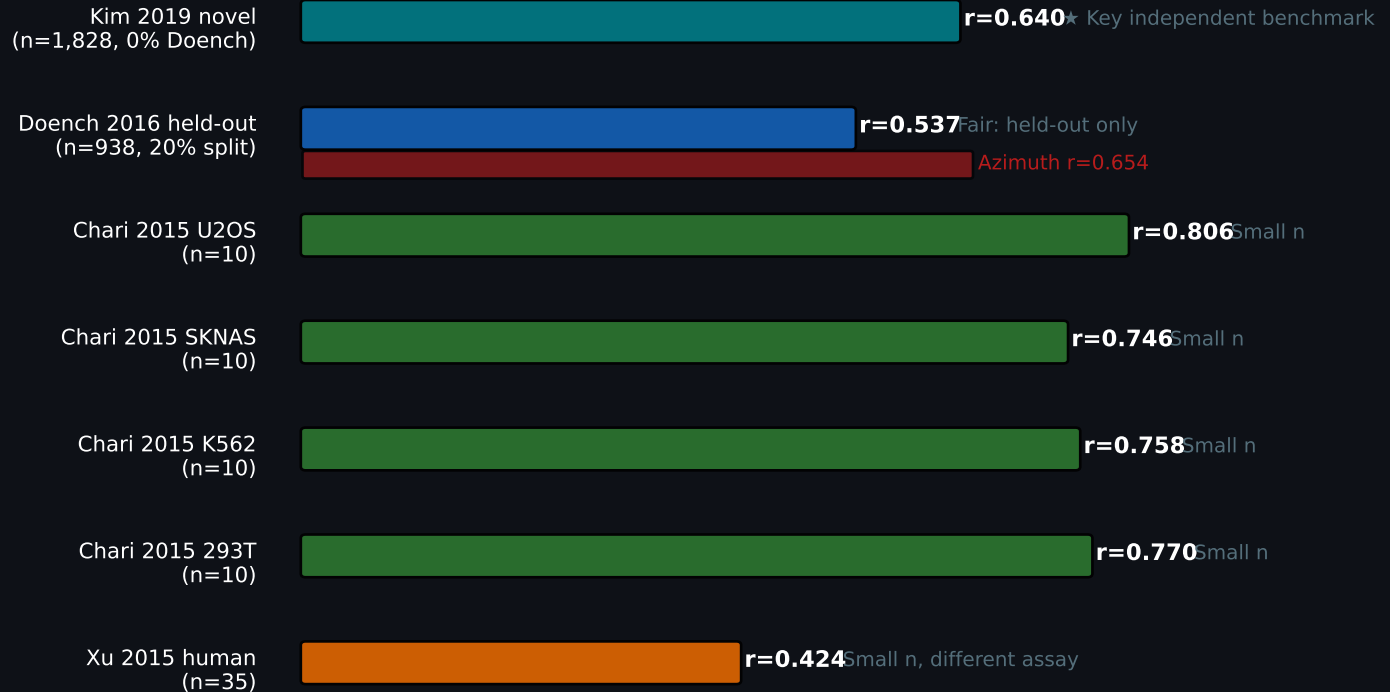
$\exp(-17^2 / (2 \times 50^2)) = \exp(-0.0578) = 0.944$

Combined ($w=0.4$)

$(1-0.4) \times 0.599 + 0.4 \times 0.944 = 0.359 + 0.378 = 0.737$

Benchmark Results

Performance on held-out and independent datasets



Precision@k — Kim 2019 (threshold = 80th percentile efficiency)
CRISPRscan zebrafish (n=1,020)

P@k = fraction of top-k predicted guides whose true efficiency exceeds the 80th percentile



Kim 2019 URL: https://raw.githubusercontent.com/L-Q-Y/CRISPRtool/main/data/Cas9/Kim2019_train.csv

Run `python compare_azimuth.py` and `python independent_validation.py` to reproduce

gRNA Predictor (ours)
Azimuth (trained on ALL Doench)

How To Use gRNA Predictor

Step-by-step guide for designing CRISPR experiments

Step 1 — Paste Your DNA Sequence

Paste a raw DNA sequence (A, C, G, T only — spaces are ignored). The tool scans both strands automatically. Minimum ~40 bp; practical range: 100 bp–10 kbp. If you have a gene of interest, paste the full exon or coding region.

Step 2 — Choose PAM Type

Select the nuclease PAM:

NGG → SpCas9 (default, most common, human cells)

NAG → SpCas9 alternative PAM (lower efficiency, more sites)

NNGRRT → SaCas9 (smaller nuclease, for AAV delivery vectors)

TTTV → Cas12a/Cpf1 (T-rich PAM, staggered DSB, 5'-PAM)

Step 3 — Optionally Set Target Position

If you want to edit at a specific location (e.g., introduce a mutation at codon 248), enter the 1-based position in the sequence. The tool will add a Gaussian proximity bonus for guides whose cut site is near your target. Default proximity weight = 0.4 (40% proximity, 60% efficiency×specificity).

Step 4 — Click PREDICT GRNAS

Results appear in a table showing the top 5 guides ranked by combined_score. For each guide you see: sequence, PAM, position, strand, efficiency score, off-target specificity, cut site, distance to target, combined score, and GC%.

Step 5 — Interpret the Results Table

combined_score: primary ranking metric (higher = better, max 1.0)

score: raw XGBoost efficiency prediction

off_target_score: specificity heuristic (1.0 = most specific)

cut_site: 1-based position where Cas9 cuts (choose for HDR templates)

distance_to_target: base pairs from cut site to target (if applicable)

GC%: guide GC content (aim for 40–60%)

Local Development Setup

```
# Backend (Python 3.13 + FastAPI)
cd backend && source ../venv/Scripts/activate
uvicorn app.main:app --reload
```

```
# Frontend (React 18 + Vite, separate terminal)
cd frontend && npm run dev
```

```
# Retrain model (after downloading fresh data)
python download_datasets.py && python train_model.py
```

gRNA Predictor v4.5 | Source: github.com/jana242k4/gRNA_Predictor